

Can We Predict SACCO Financial Surplus?

A Machine Learning Approach to Financial Forecasting for Kenyan SACCOs

Phase 5 Group Project | Group 5 | github.com/moruriinnocentlouis-collab/phase_5_project_group_5

52

SACCOs

2007–2023

Years of Data

9

Models Tested

82%

Best Accuracy

The Problem — and Why It Matters

The Challenge

SACCOs are Kenya's primary source of affordable credit for millions of members. Yet most SACCO managers have no reliable way to forecast whether the organisation will end the year with a surplus or a deficit.

Without this forecast, decisions on dividends, loan limits, and reserve allocation are made largely on instinct.

Our question:

Can 10 standard balance sheet numbers predict a SACCO's surplus for the year?

Why It Matters

- Plan dividends before year-end
- Set safe loan portfolio limits
- Meet SASRA capital requirements
- Spot underperforming SACCOs early

The Data — 52 SACCOs, 16 Years

What We Used

Source	SASRA annual supervision reports
Scope	52 deposit-taking SACCOs
Period	2007 to 2023 (16 years)
Target	Current Year's Surplus
Format	One CSV per SACCO in data_clean/

10 Input Numbers

- Total Assets
- Gross Loan Portfolio
- Total Liabilities
- Total Deposit Liabilities
- Total Equity
- Net Loan Portfolio
- Financial Investments
- Cash & Cash Equivalent
- Share Capital
- Statutory Reserve

Data Cleaning: Data reshaped from wide to long format for analysis.
Zero values treated as missing (NaN).

What the Data Tells Us

01 Sector Growth

Performance was flat for a decade, then nearly tripled between 2019 and 2022. Stima SACCO holds significantly more value than any other institution.

02 It Is a Loan Business

Total Assets, Gross Loans, and Total Liabilities move almost in lockstep. The sector's core activity is lending — deposits fund loans almost directly.

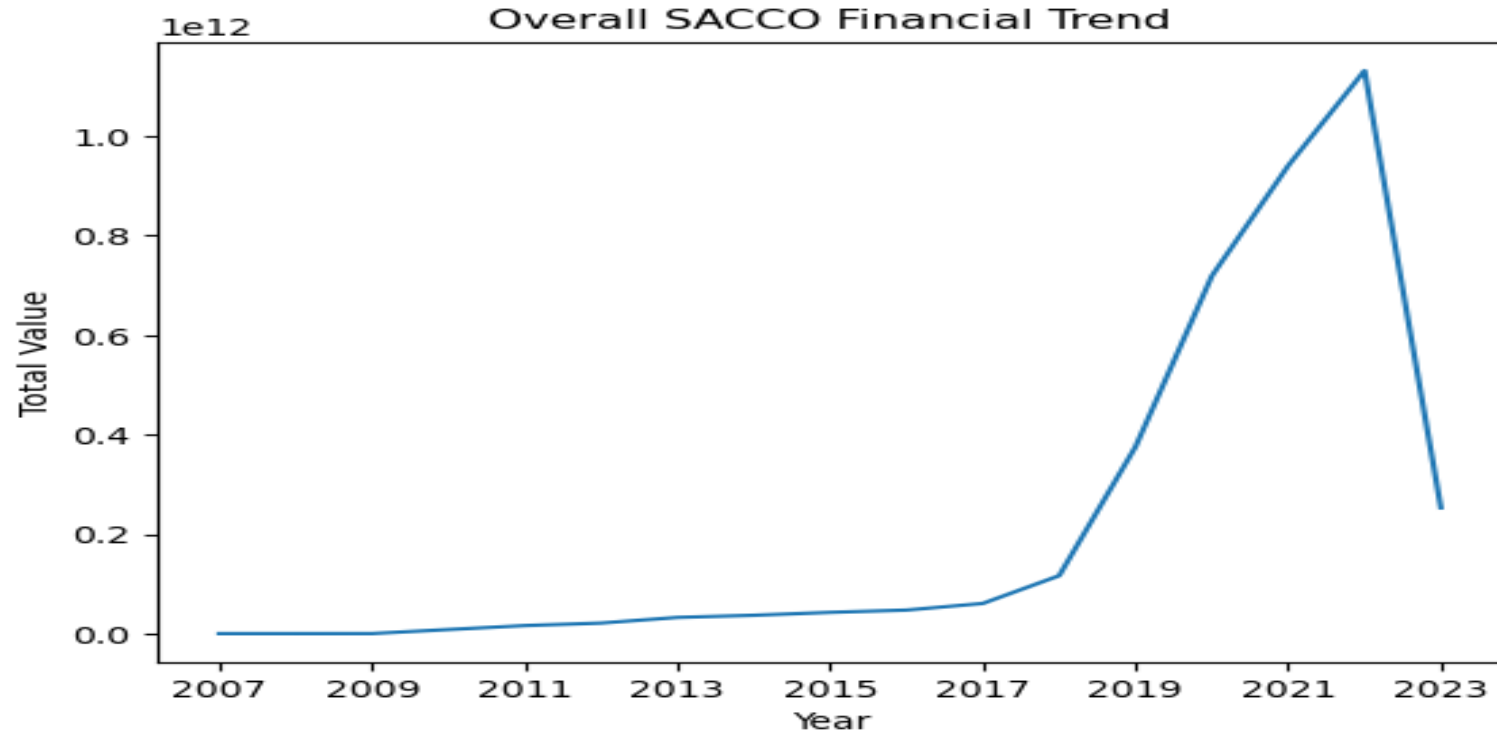
03 A Growing Funding Gap

Since 2018, loans have grown faster than deposits. By 2023 loans reached ~KES 5 billion while deposits sat at ~KES 4.5 billion — a sign of stretched liquidity.

04 Capital Structure Matters Most

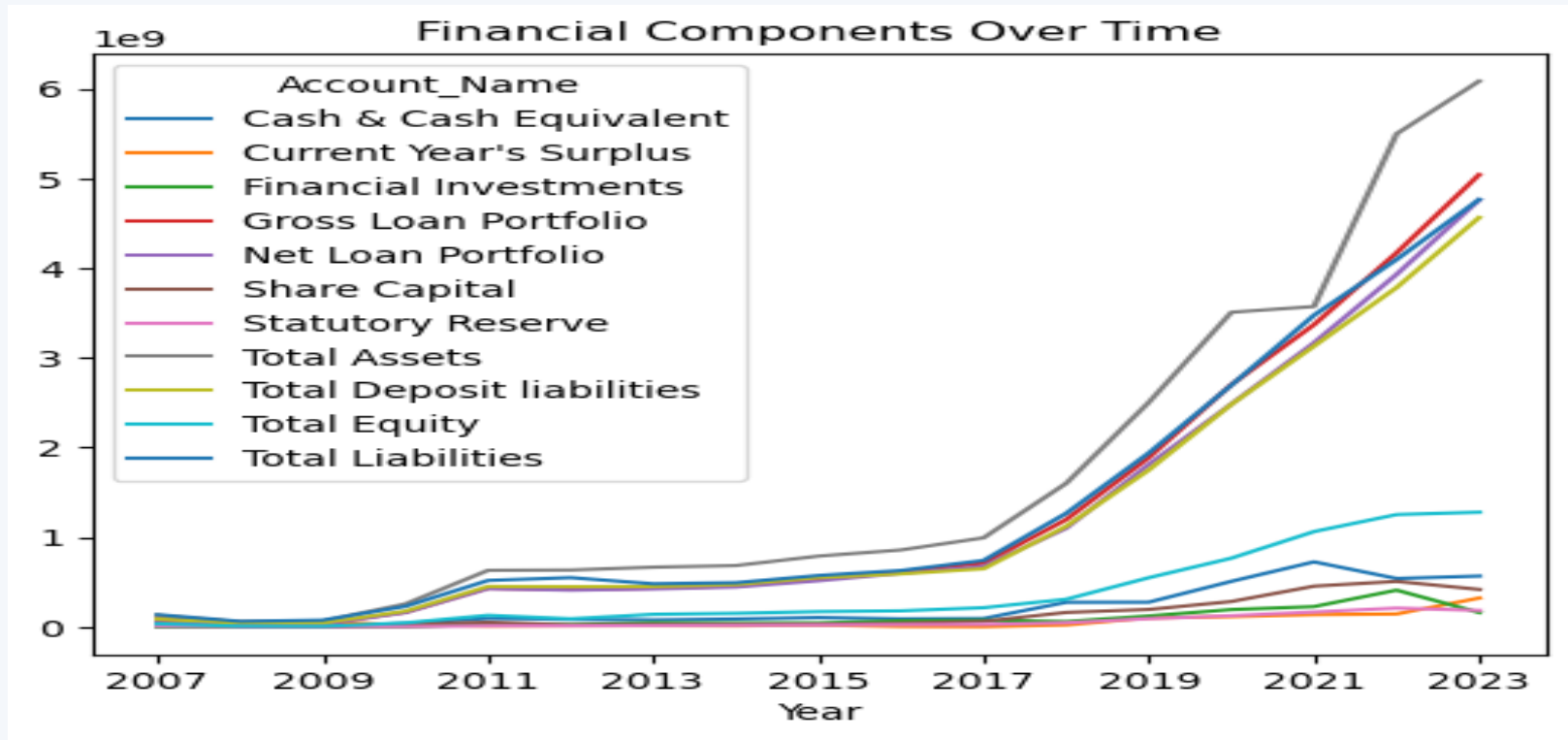
Across all models, Total Liabilities and Total Equity were the top predictors of surplus — not loan size. How a SACCO is financed matters more than how much it lends.

Overall SACCO Financial Trend



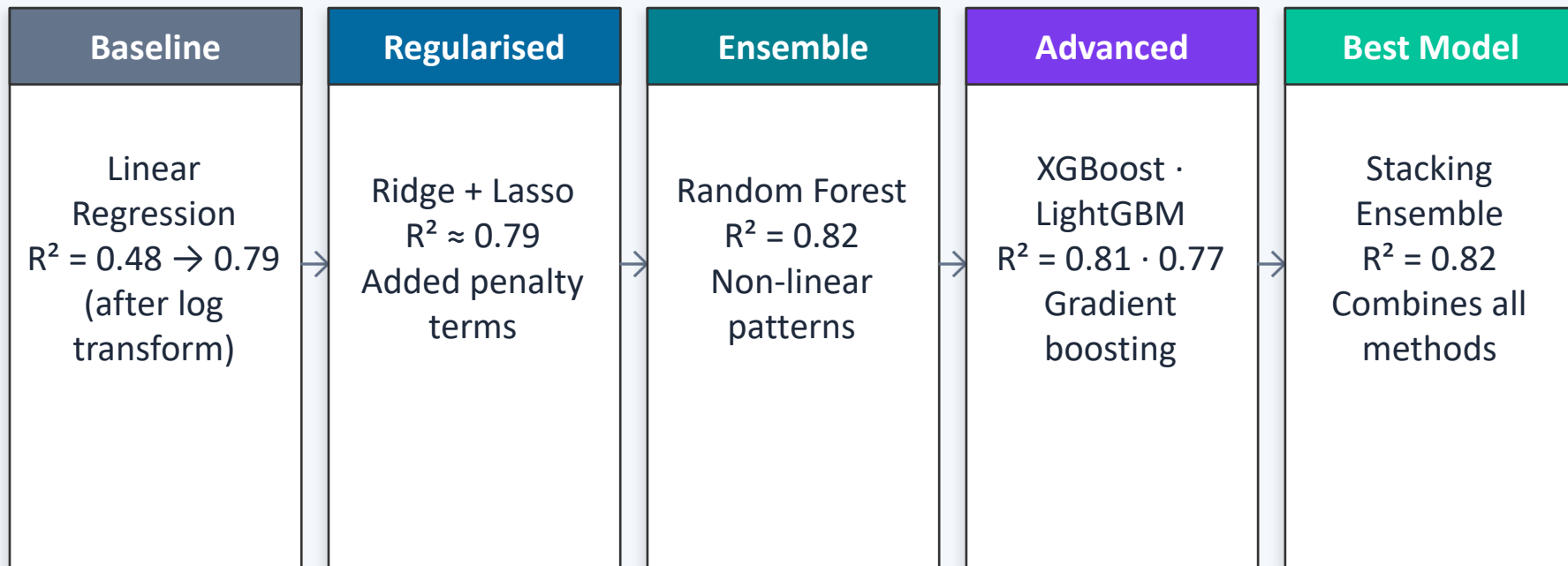
Performance was stable 2007–2016, grew gradually 2017–2019, then surged sharply 2019–2022. The 2023 decline likely reflects incomplete data or external economic factors. Strong long-term growth overall.

Financial Components Over Time



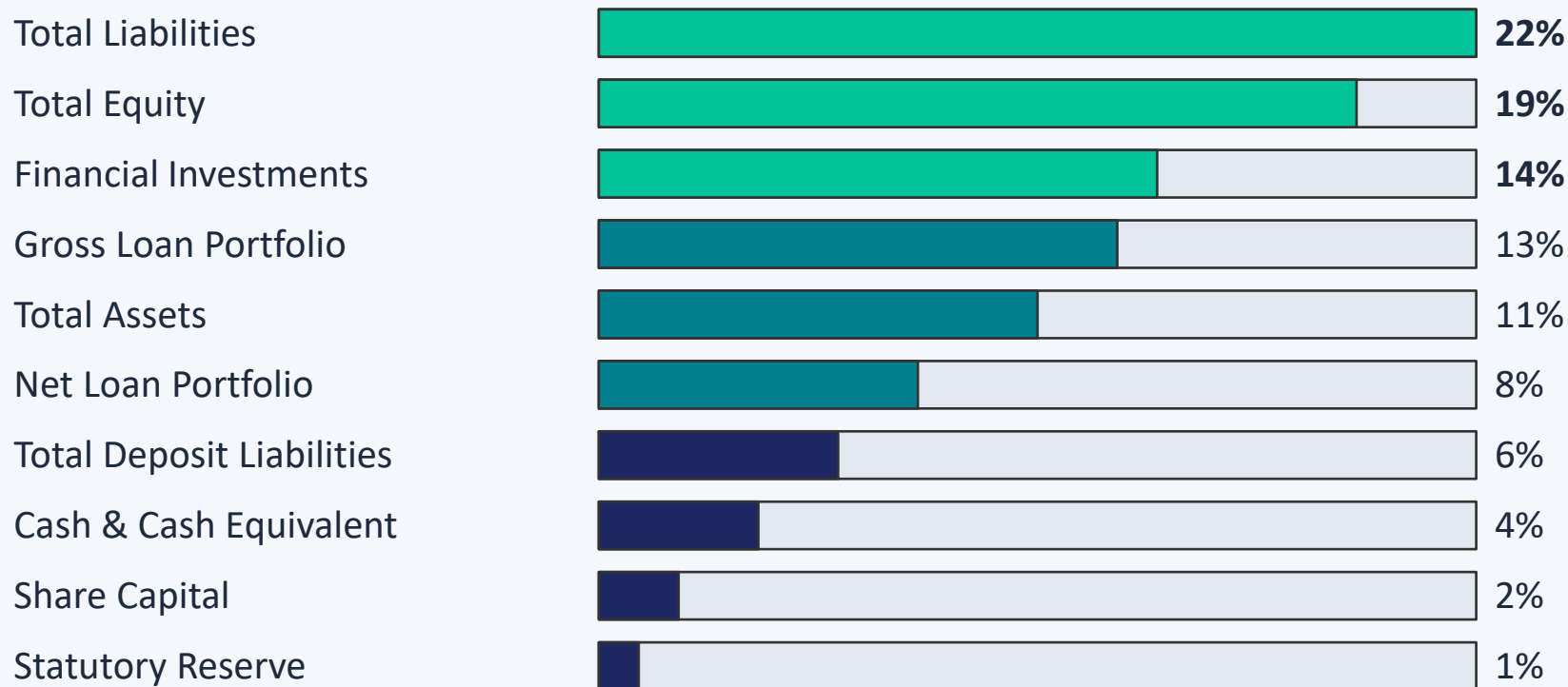
Total Assets leads growth from 2017. Loans, liabilities, and deposits move in tight synchronisation. The 2017–2018 inflection point marks the sector's period of aggressive expansion.

How We Built the Models — From Simple to Advanced



Why log transform? Financial numbers span billions to millions — we scaled them so models could compare them fairly. This alone boosted accuracy from 48% to 79%.

What Drives a SACCO's Surplus?



It is not about how much a SACCO lends. It is about how it is financed. Liabilities and equity structure predict surplus best.

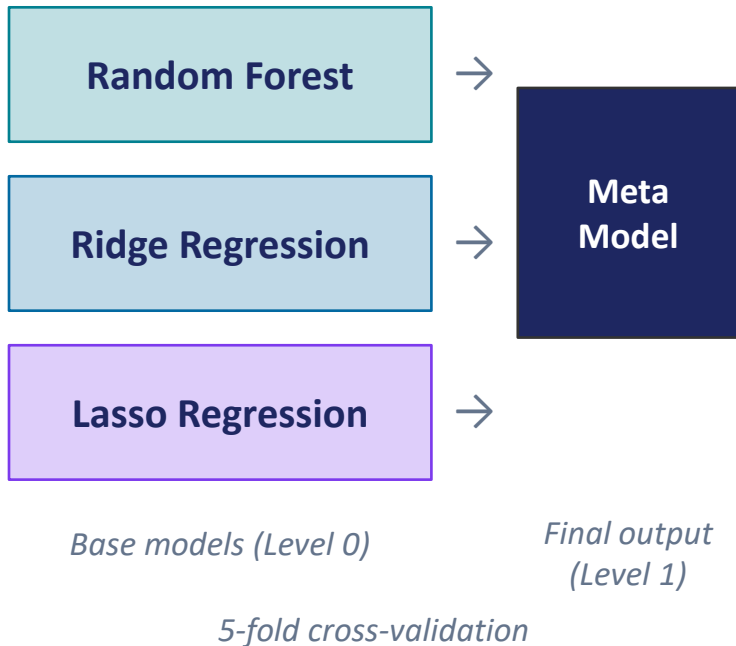
How All 9 Models Compared

Model	Category	Accuracy (R^2)	Result
Linear Regression (basic)	Starting point	0.48	Weak
Linear Regression (log-scaled)	Starting point	0.79	Good after scaling
Ridge Regression	Regularised	0.79	Same as above
Lasso Regression	Regularised	0.78	Slightly lower
Random Forest	Ensemble	0.82	Strong
XGBoost	Ensemble	0.81	Strong
LightGBM	Ensemble	0.77	Less consistent
Neural Network	Deep learning	0.76	Competitive but slow
Stacking Ensemble	Best model	0.82	Winner — combines all

R^2 = how much of the variation in surplus the model correctly explains. 0.82 = 82%.

Our Best Model — The Stacking Ensemble

How It Works



What Is Stacking?

Think of it like hiring three analysts and then a senior manager.

The three analysts (Random Forest, Ridge, and Lasso) each make their own surplus prediction independently.

The senior manager (a Linear Regression meta-model) looks at all three predictions and learns which analyst to trust in which situation.

The result is smarter than any single analyst working alone.

$R^2 = 0.82$ — explains 82% of variation in SACCO surplus

Why We Chose the Stacking Ensemble

Model	R ²	Handles non-linear patterns	Stable with small datasets	Chosen?
Linear Regression	0.79	No	Yes	No
Random Forest	0.82	Yes	Moderate	No
XGBoost	0.81	Yes	Moderate	No
Neural Network	0.76	Yes	No	No
Stacking Ensemble	0.82	Yes	Yes	Yes

1 Best Accuracy

$R^2 = 0.82$ — the highest score across all 9 models tested. It matches Random Forest but does so more consistently by combining multiple perspectives.

2 Uses Strengths of Each Model

Random Forest finds non-linear patterns. Ridge keeps the model stable. Lasso filters out weak features. The meta-model decides how much to rely on each.

3 Handles SACCO Data Well

SACCO financials are messy — skewed values, varied sizes, missing data. A stacking approach is more robust to these quirks than any single algorithm.

From Model to Tool — The Streamlit App

What the App Does

Model Training

Retrain any of the 9 models, compare results, and download the model file.

Predict Surplus

Enter 10 balance sheet numbers and get an instant surplus prediction.

How to Access

- 1 Clone the GitHub repo
- 2 `pip install -r requirements.txt`
- 3 `streamlit run app.py`
- 4 Open browser at localhost:8501

Also hosted at share.streamlit.io — no installation needed

What We Found

- 1 Yes — you can predict surplus from 10 balance sheet numbers with 82% accuracy.
- 2 The Stacking Ensemble wins because it combines three different models, letting each contribute its strengths.
- 3 How a SACCO is financed (liabilities and equity) predicts its surplus better than how much it lends.
- 4 A simple web app now puts this prediction tool in the hands of any SACCO manager.